

DS-5343

How is visualized

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Índice



1. Recursos Visuales
2. Canales para cada tipo de dato
3. Expresividad
4. Efectividad



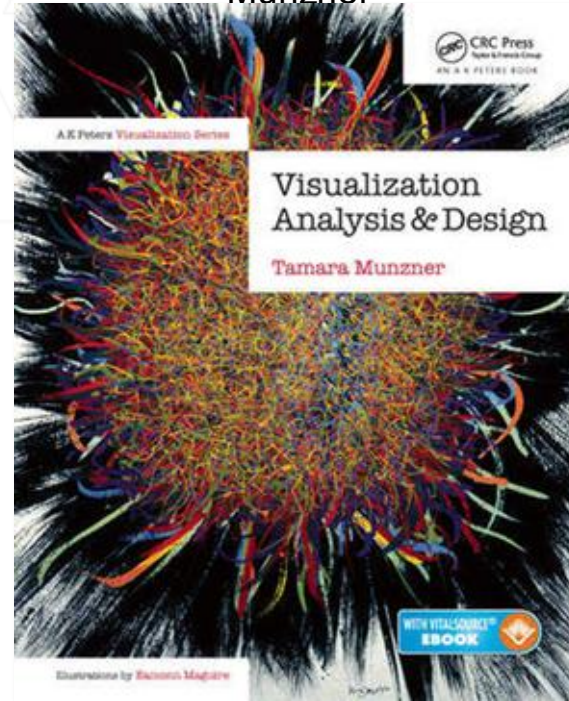
Objetivos de la Sesión

En esta sesión, se explorarán los **principios fundamentales de la visualización de datos**, centrándose en el uso de **marcas y canales visuales** para representar información de manera efectiva. Se analizarán los tipos de canales según su naturaleza ordenada o categórica, así como los conceptos de **expresividad y efectividad** en el diseño de visualizaciones. Al final de la sesión, los estudiantes comprenderán cómo seleccionar las mejores estrategias visuales para comunicar datos de forma clara y precisa.



Libro

Visualization Analysis and Design, Tamara Munzner



1.

Elementos gráficos y Canales



How?

Encode

→ Arrange

→ Express



→ Separate



→ Order



→ Align



→ Use



→ Map

from **categorical** and **ordered** attributes

→ Color

→ Hue



→ Saturation



→ Luminance



→ Size, Angle, Curvature, ...



→ Shape



→ Motion

Direction, Rate, Frequency, ...



Manipulate

→ Change



→ Select

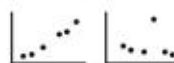


→ Navigate



Facet

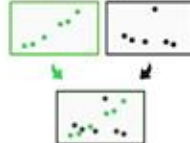
→ Juxtapose



→ Partition



→ Superimpose



Reduce

→ Filter



→ Aggregate



→ Embed

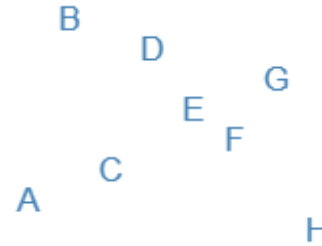
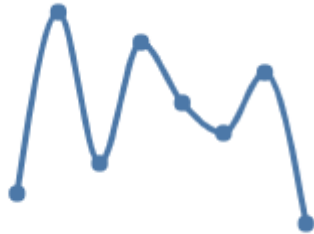
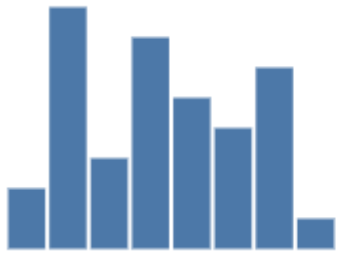


What?

Why?

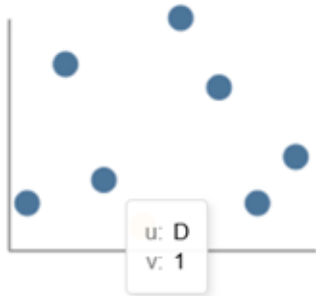
How?

ANALIZAR LA ESTRUCTURA DE LA VISUALIZACIÓN

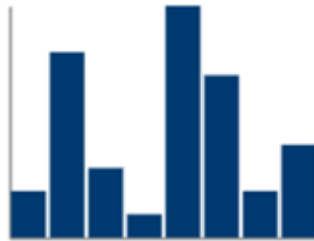


MARKS

Point



Line



Area



CANALES

→ Position

→ Horizontal



→ Vertical



→ Both



→ Color



→ Shape



→ Tilt



→ Size

→ Length



→ Area



→ Volume



TIPO DE CANALES

Channels: Expressiveness Types and Effectiveness Ranks

➔ **Magnitude Channels: Ordered Attributes**

Position on common scale



Position on unaligned scale



Length (1D size)



Tilt/angle



Area (2D size)



Depth (3D position)



Color luminance



Color saturation



Curvature



Volume (3D size)



➔ **Identity Channels: Categorical Attributes**

Spatial region



Color hue



Motion



Shape



Most

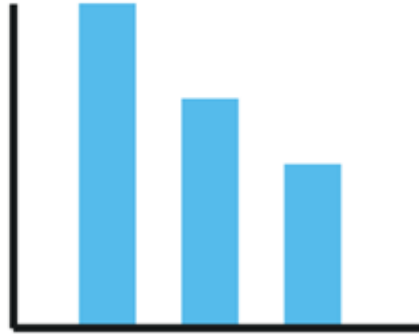
Effectiveness

Least

Same

Same

¿QUÉ MARCA Y CANAL?



¿QUÉ MARCA Y CANAL?



¿QUÉ MARCA Y CANAL?



¿QUÉ MARCA Y CANAL?



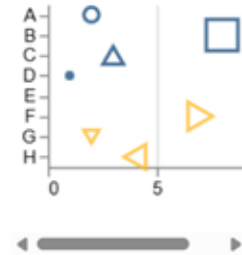
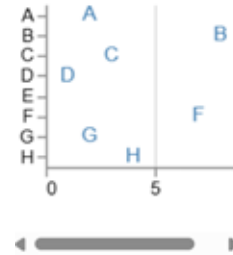
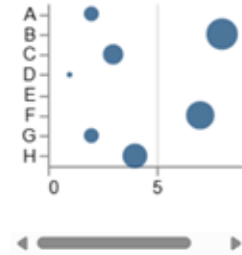
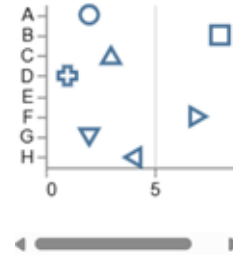
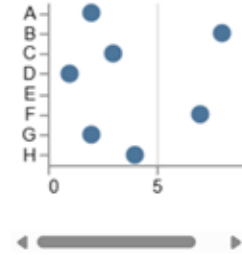
2.

Ejemplos



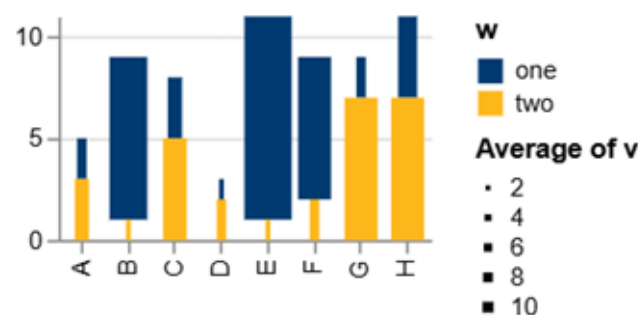
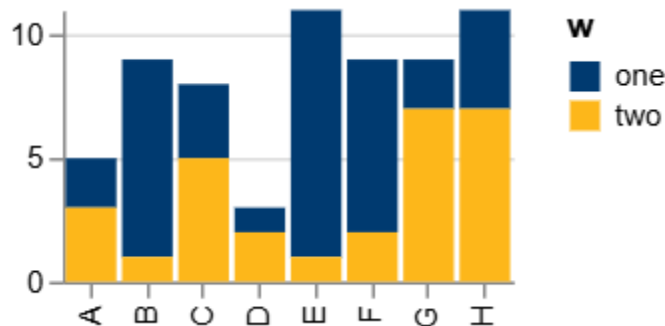
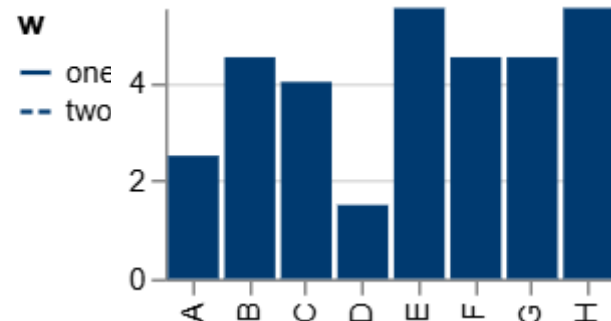
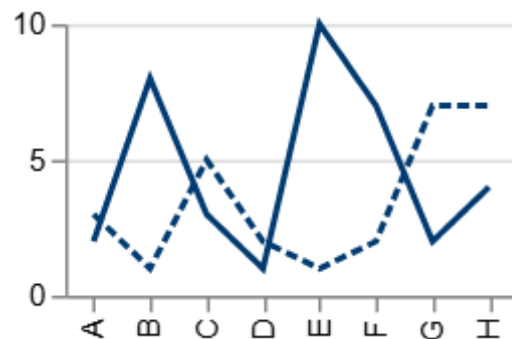
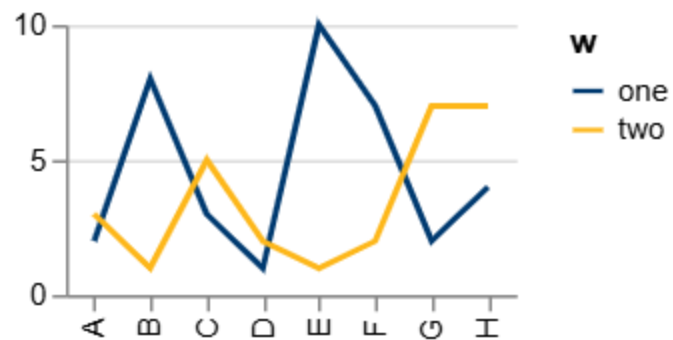
PUNTOS

- Zero-dimensional
- Transmite posición
- Puede ser codificado por tamaño y forma



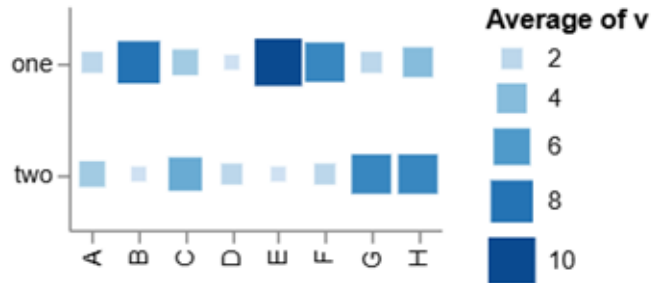
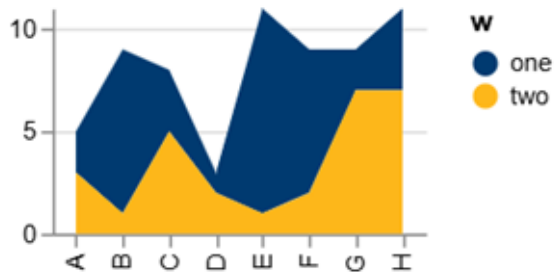
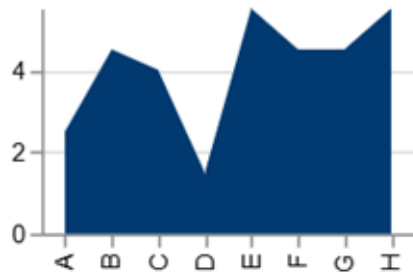
LINEAS

- One-dimensional
- Transmite posición y Longitud
- Se puede codificar Ancho



ÁREAS

- Two-dimensional
- Completamente restringidos



VARIABLES OF THE IMAGE

XY 2 DIMENSIONS OF THE PLANE

Z

SIZE

VALUE

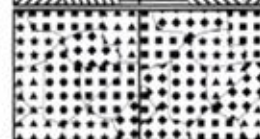
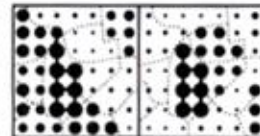
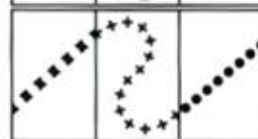
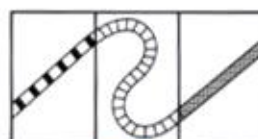
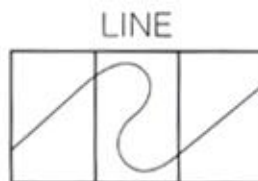
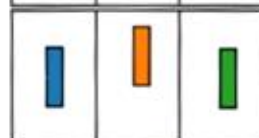
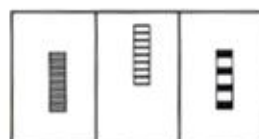
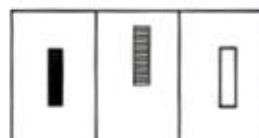
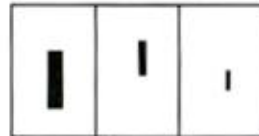
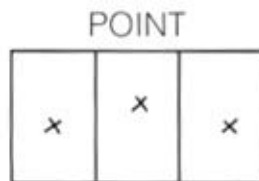
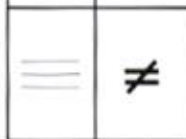
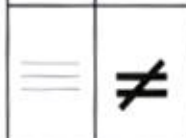
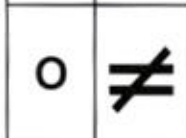
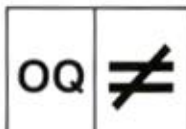
DIFFERENTIAL VARIABLES

TEXTURE

COLOR

ORIENTATION

SHAPE



2.

Efectividad y Expresividad



EXPRESIVIDAD

- La codificación visual debe expresar toda y únicamente la información contenida en el conjunto de datos.
- Los datos ordenados deben representarse de una manera que percibamos como ordenada.
- Debe haber una correspondencia entre las características del canal visual y las del tipo de datos.

EFFECTIVIDAD

Codifique los atributos más importantes con los canales de mayor clasificación

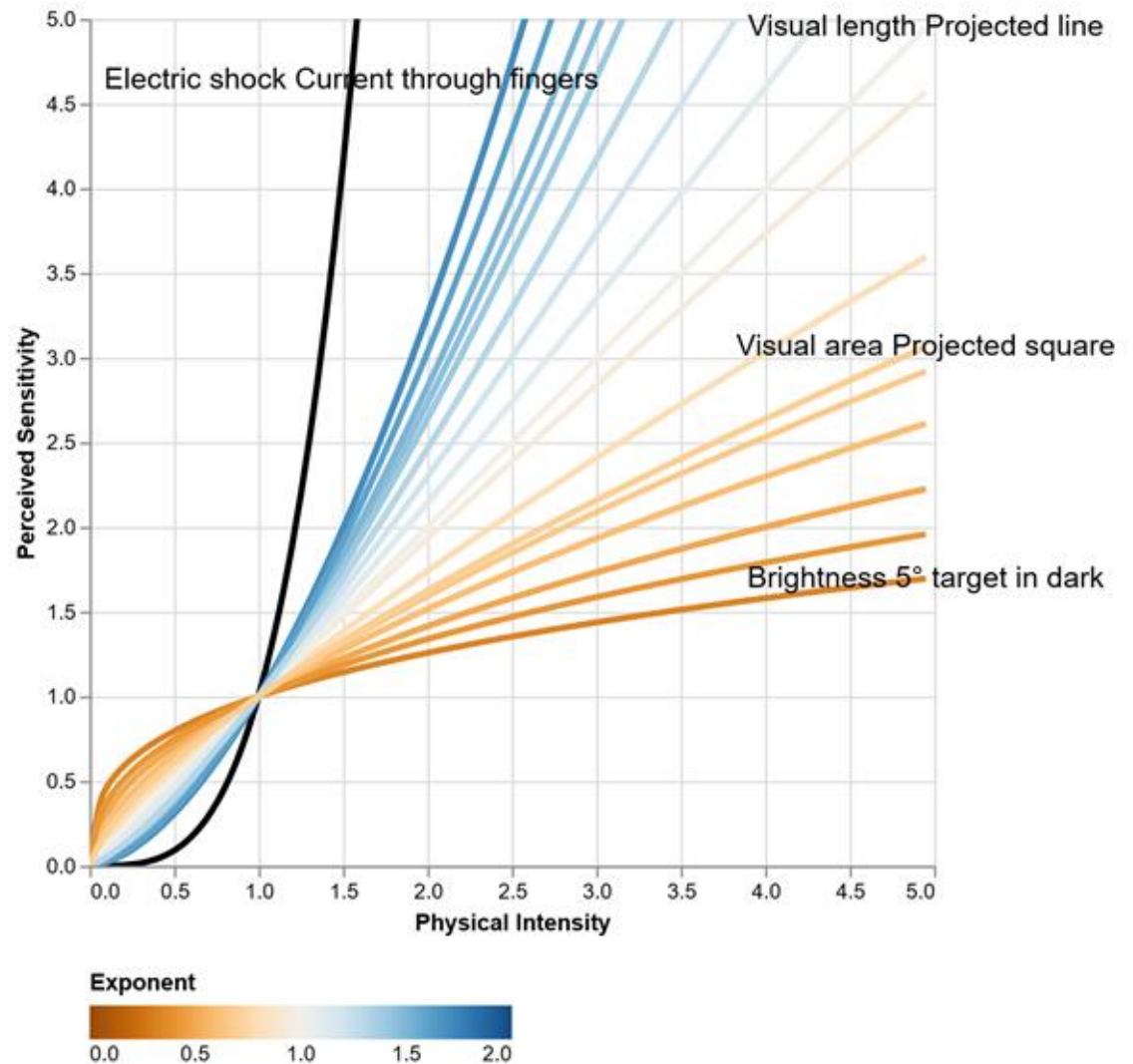
¿De qué depende el ranking?

¿DE QUÉ DEPENDE EL RANKING?

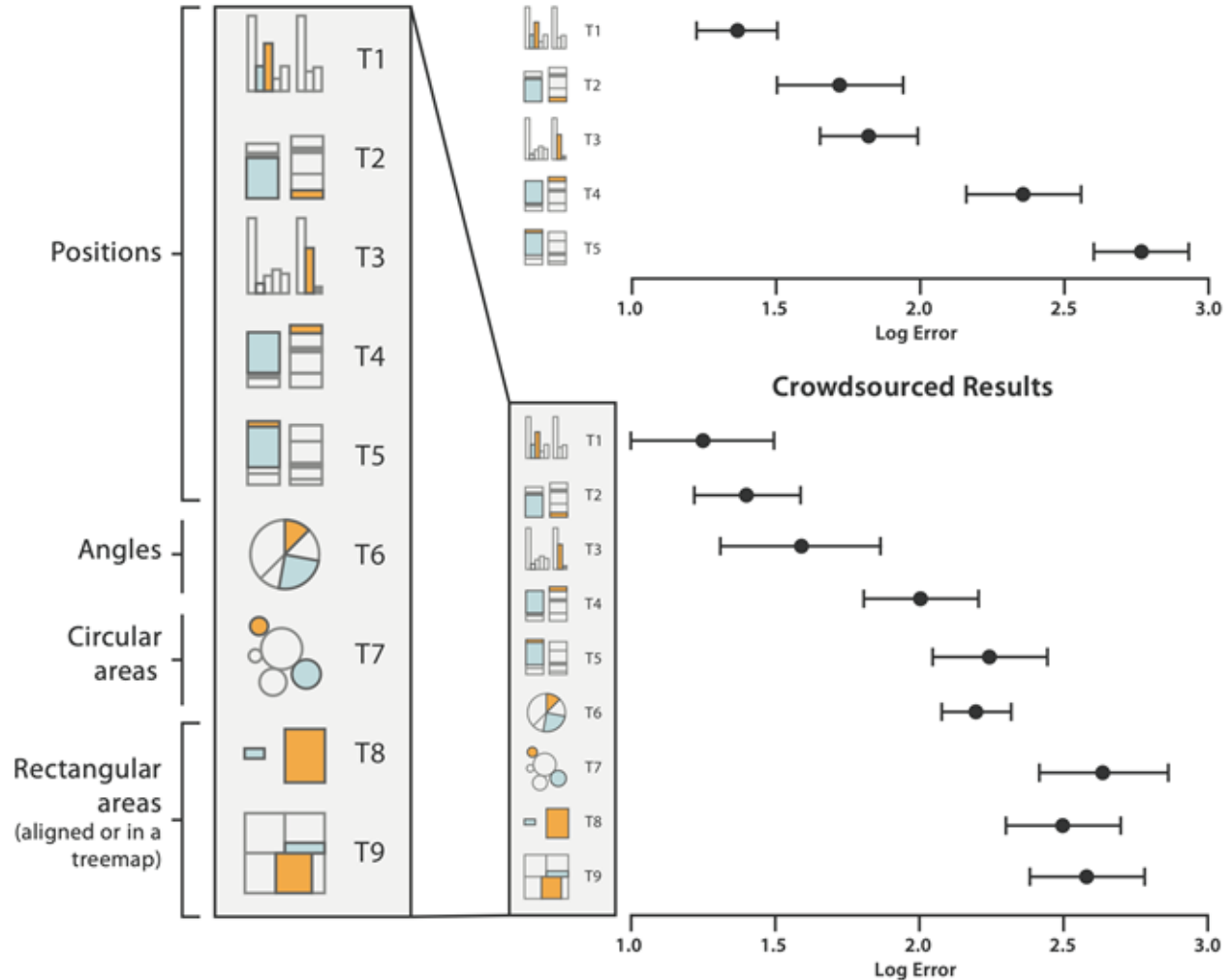
- Precisión
- Capacidad de discriminación
- Separabilidad
- Destacabilidad (Popout)

EFFECTIVIDAD (TEORÍA)

- Tamaño
- Color
- Longitud



EFFECTIVIDAD (EXPERIMENTOS)



[Crowdsourcing Graphical Perception: Using MechanicalTurk to Assess Visualization Design](#)

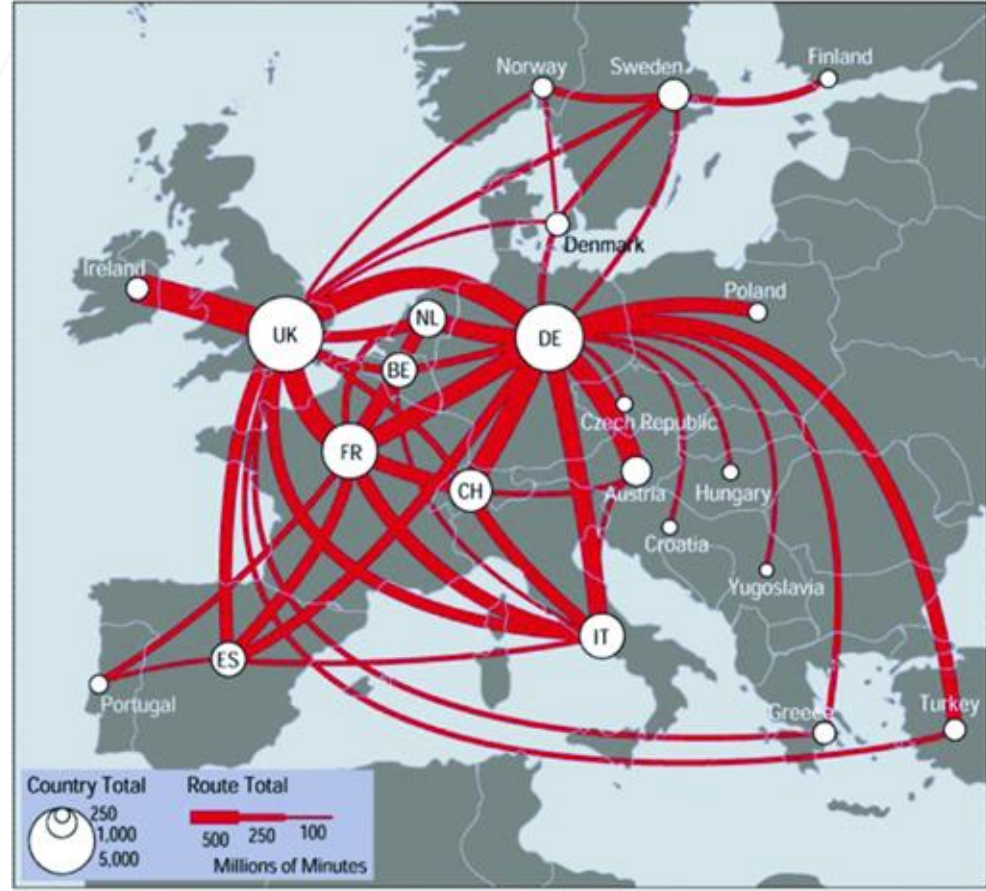
BUSCA DE EFECTIVIDAD: Discriminación

¿Cuántos pasos son utilizables?

Ancho de línea, no muchos

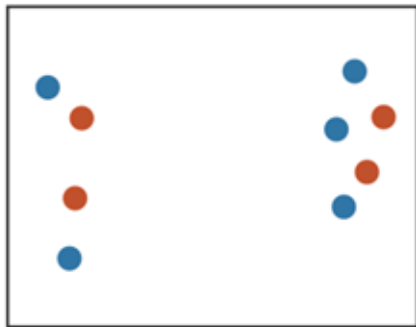
scientific visualization altitude

Mappa.mundi.net TeleGeography's Traffic Flow



BUSCA DE EFECTIVIDAD: Separación

Position
+ Hue (Color)



Fully separable

Size
+ Hue (Color)



Some interference

Width
+ Height



Some/significant
interference

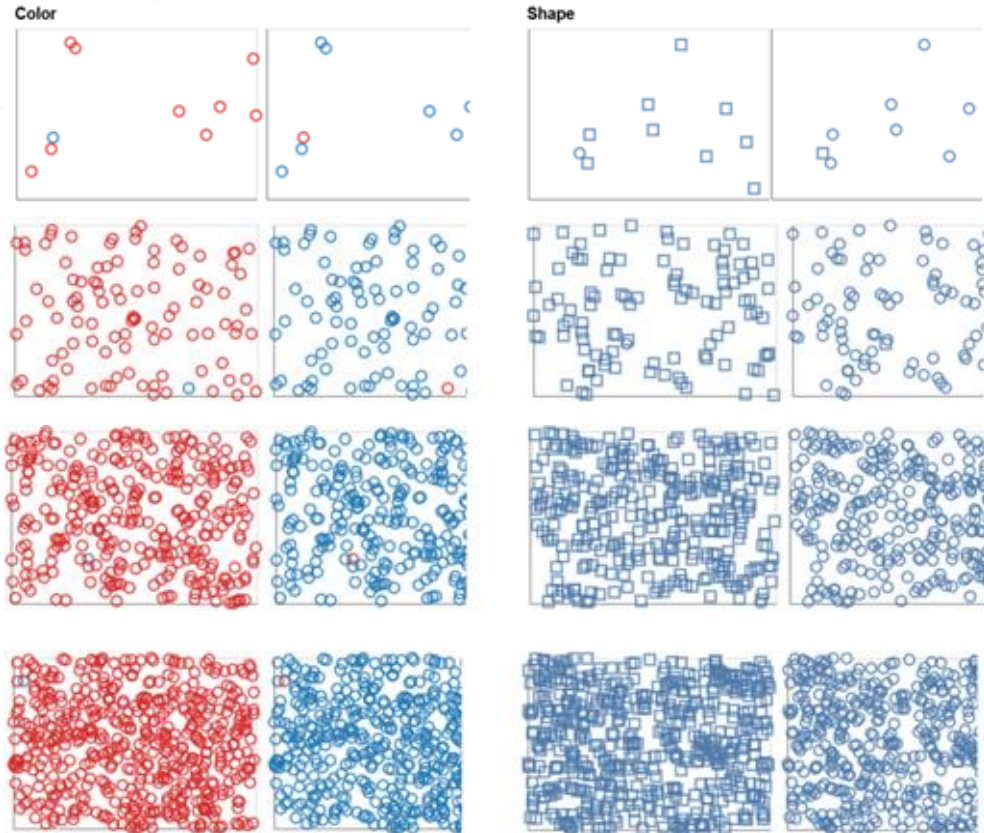
Red
+ Green

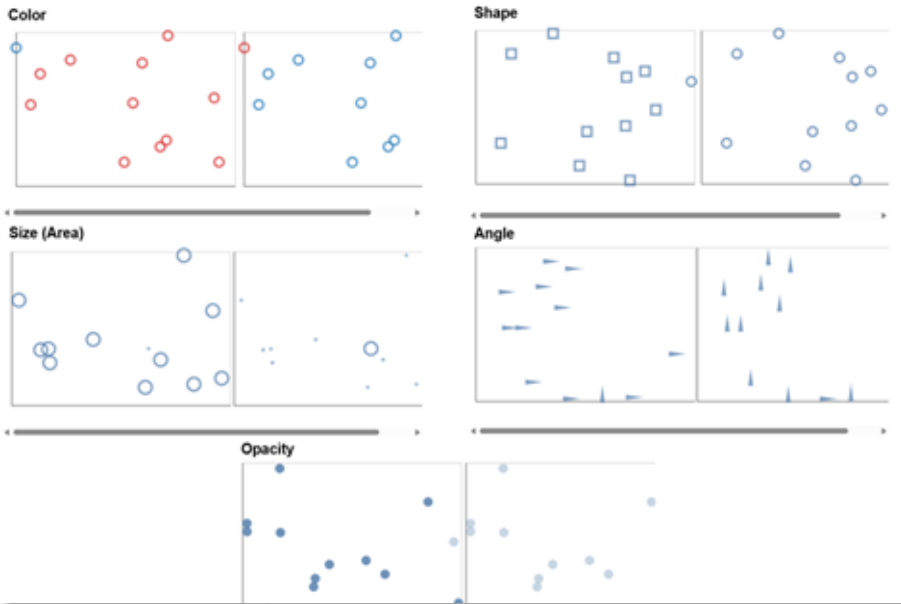


Major interference

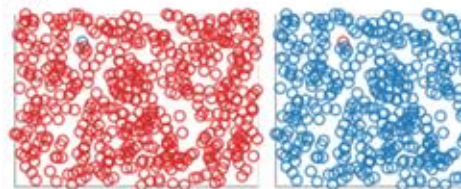
BUSCA DE EFECTIVIDAD: Popout

- Encuentra los elementos que no encajan.
- ¿Cuáles resaltan? (Popout)





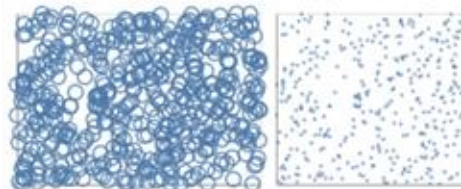
Color



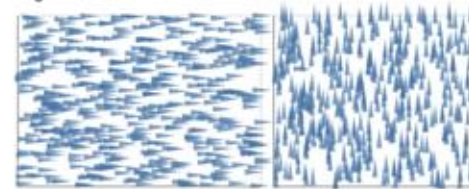
Shape



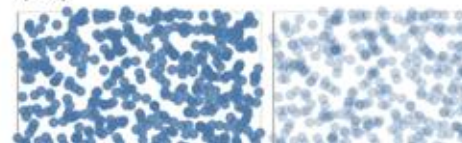
Size (Area)



Angle



Opacity



BUSCA DE EFECTIVIDAD: Agrupamiento

Marks as Links

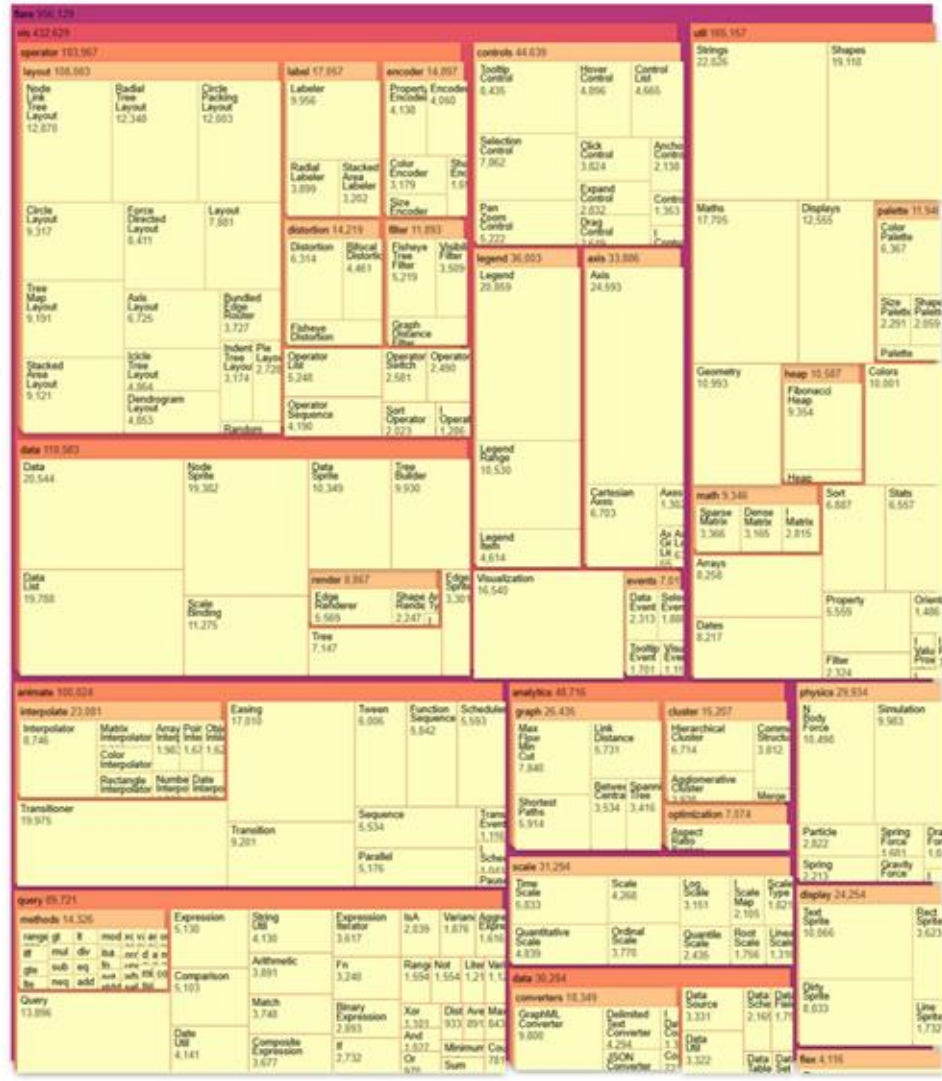
➔ Containment



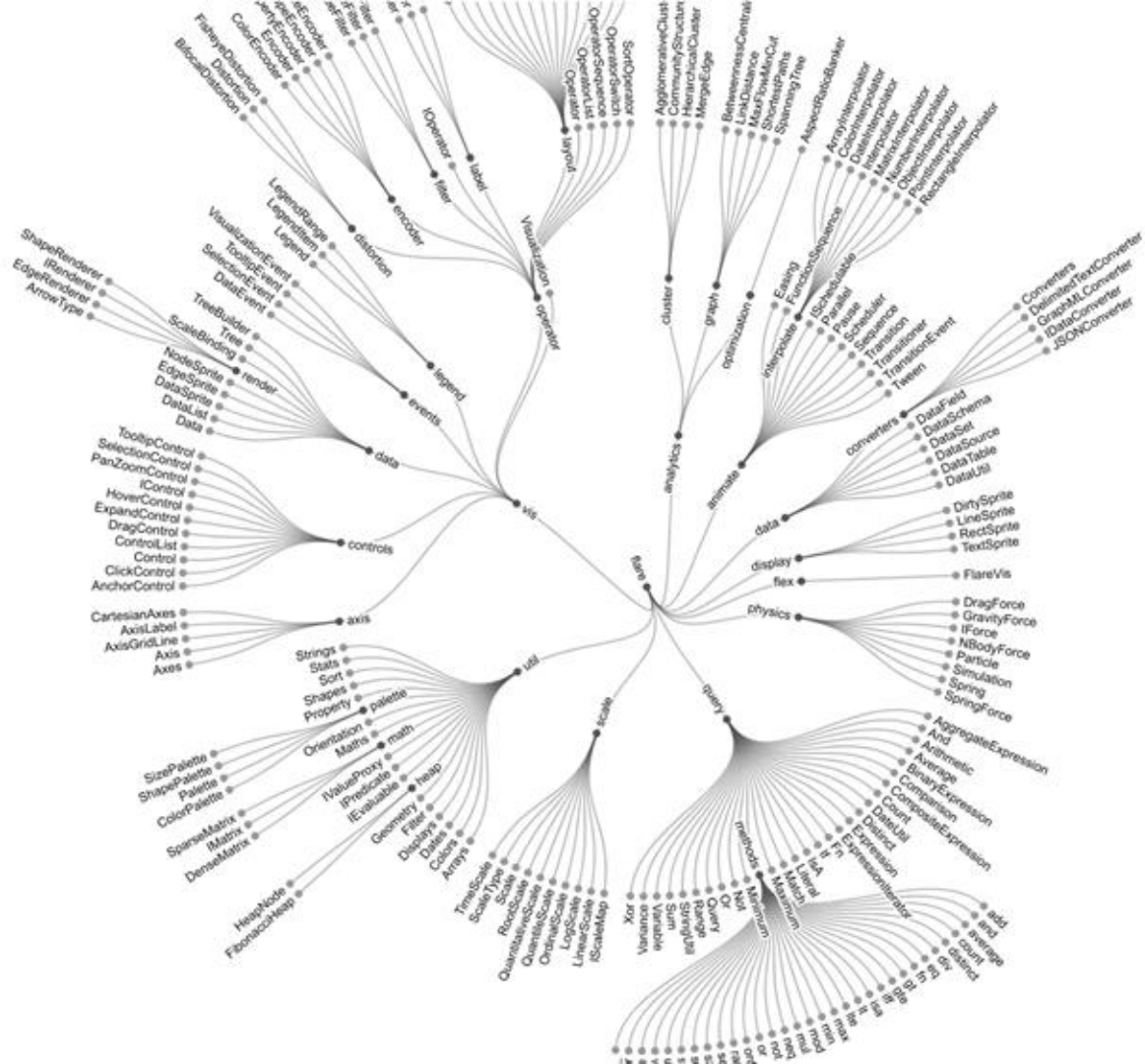
➔ Connection



Containment



Connection example



GROUPING

- Proximidad 🖱️ usar separación Espacial
- Similaridad 🖱️ Usar canales categóricos

➔ Identity Channels: Categorical Attributes

Spatial region



Color hue



Motion



Shape



☒ useGroupInABox

Enable the forceInABox

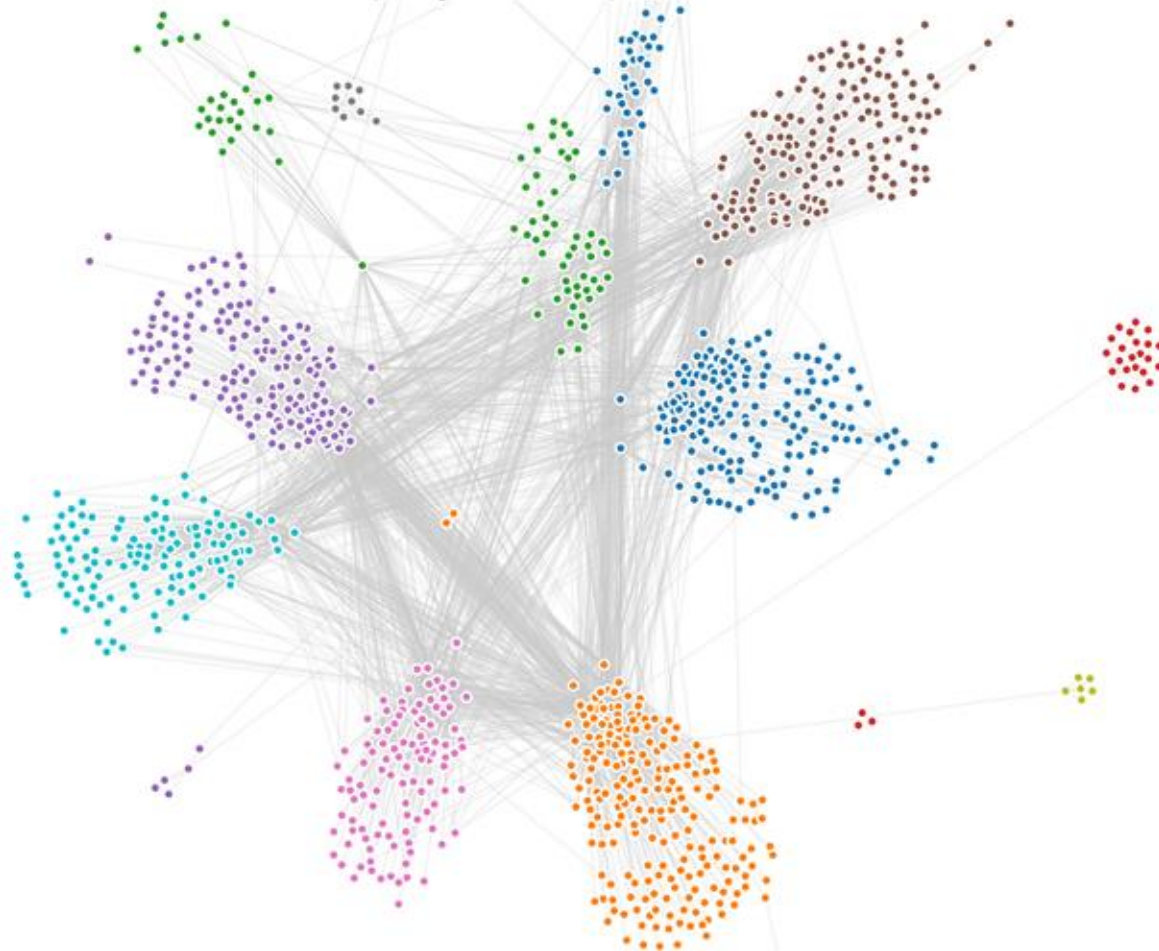
Template

force

☐ drawTemplate

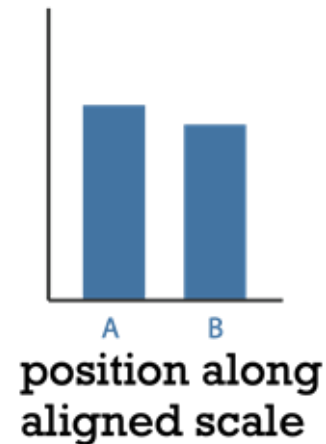
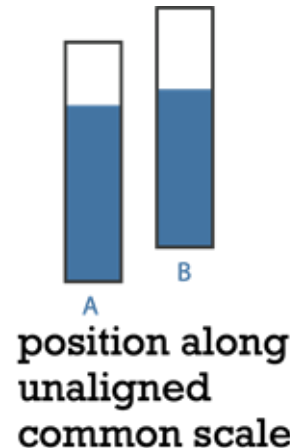
Draw the template?

The layout algorithm used to separate the clusters



JUICIOS RELATIVOS VS. ABSOLUTOS

- Percepción 👉 principalmente juicios relativos, no absolutos
- Por eso la precisión aumenta con:
 - Frame común
 - Escala y alineación



How?

Encode

→ Arrange

→ Express



→ Separate



→ Order



→ Align



→ Use



What?

Why?

How?

→ Map

from **categorical** and **ordered** attributes

→ Color

→ Hue



→ Saturation



→ Luminance



→ Size, Angle, Curvature, ...



→ Shape



→ Motion

Direction, Rate, Frequency, ...



Manipulate

→ Change



→ Select

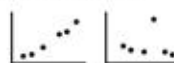


→ Navigate



Facet

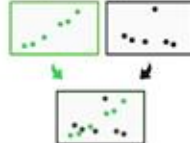
→ Juxtapose



→ Partition



→ Superimpose



Reduce

→ Filter



→ Aggregate



→ Embed



InteractiVenn: a web-based tool for the analysis of sets through Venn diagrams

InteractiVenn (Unions by list)

Unions by tree

Unions by list

Citation

Contact

Help

Terms & Privacy

List code:

Start

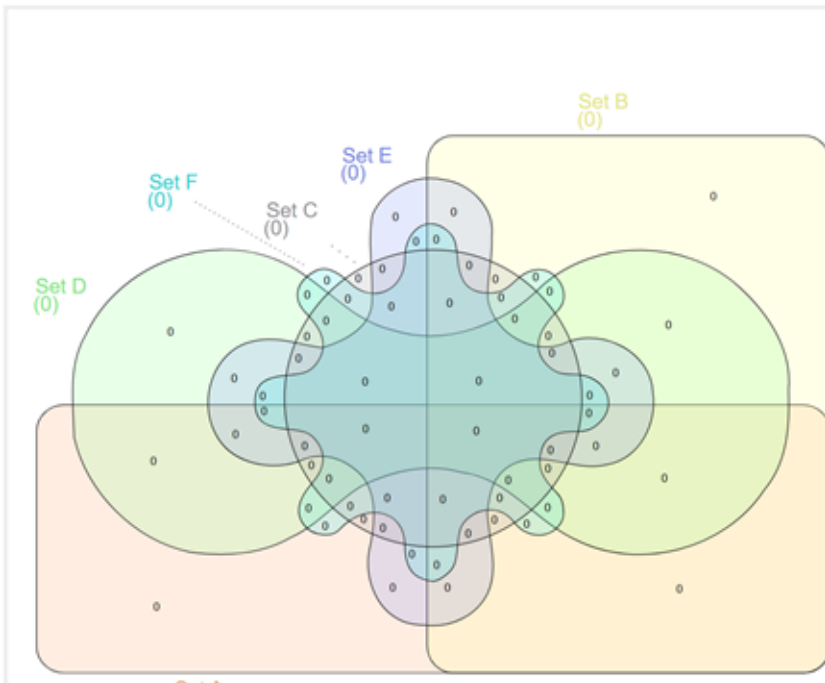
<

>

☐ Show Percentages instead (new feature!)
Mouse-over numbers highlight their sets (new feature!)

Number of Sets:

☐ 2 ☐ 3 ☐ 4 ☐ 5 ☒ 6



A: size: 0

B: size: 0

C: size: 0

D: size: 0

E: size: 0

F: size: 0

<https://observablehq.com/d/2e247c925b554494>

<https://observablehq.com/@stanfordvis/data-types-graphical-marks-and-visual-encoding-channels>

Blooket

VIZ-How



Public

From Paper to HOW (Visual Encoding)

“Understand how data is visually represented

Goal

Understand *how data is visually represented* in a research paper.

You will:

- Analyze a visualization
- Identify **marks and channels**
- Evaluate expressiveness & effectiveness

From Paper to HOW (Visual Encoding)

The HOW determines if a visualization works or fails.”:

- Same data + same task $\neq$ same visualization
- Design decisions affect understanding
- Poor encoding = misleading insights

Select **1 research paper** (IEEE VIS, EuroVis, TVCG, etc.)

Then analyze:

1. Marks
2. Visual channels
3. Mapping (data $\rightarrow$ visual)
4. Evaluate effectiveness
5. Short critique

During this class:

1. Search for a visualization research paper
2. Select **one paper**
3. **Show it to the professor for approval**



You must get approval before continuing



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Y TECNOLOGÍA

